



DIGITAL DESIGN



FINAL SYSTEM



Under supervision of:

Eng. Ali Eltemsah

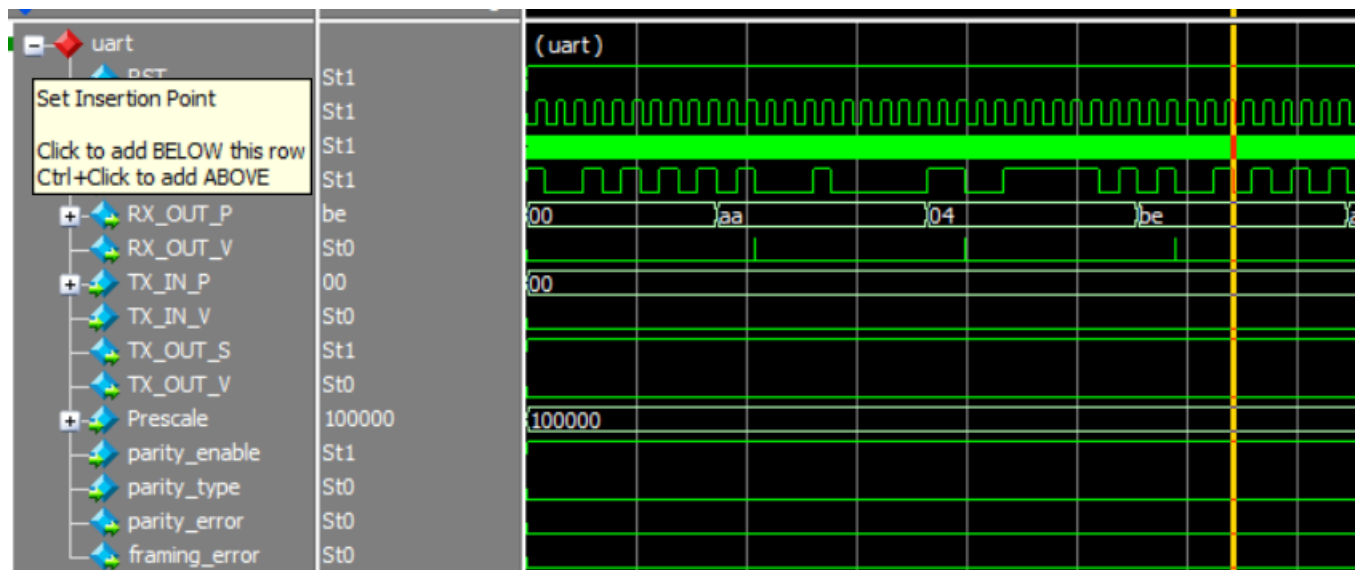
Submitted by: Mina Mikhaeel Fathy

First case : write into the register file .

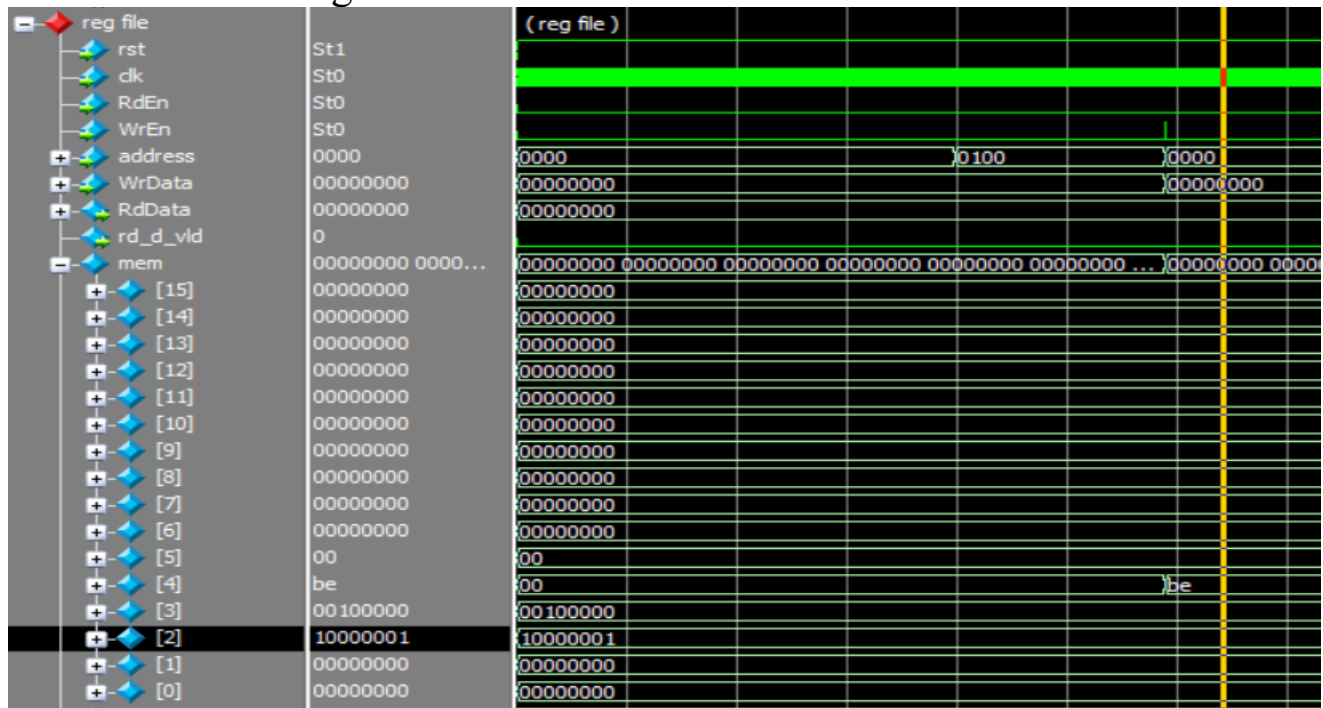
1. Register File Write command (3 frames)



Rx received frame 'aa' then the target address in the register file , then the data itself which in this case 'be' in hexadecimal .



Data received in register file address 4

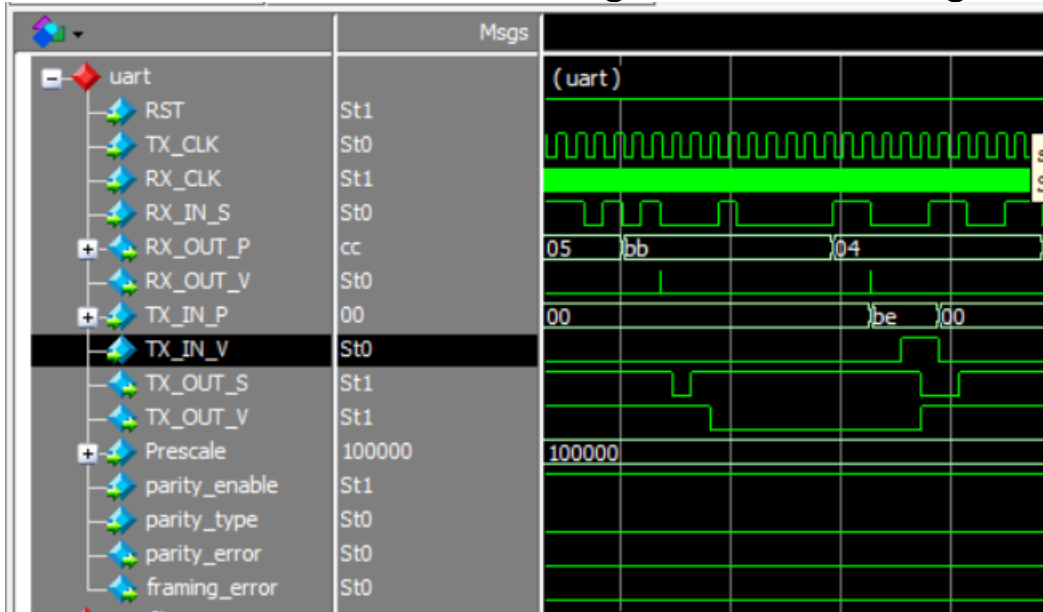


second case : read from the register file .

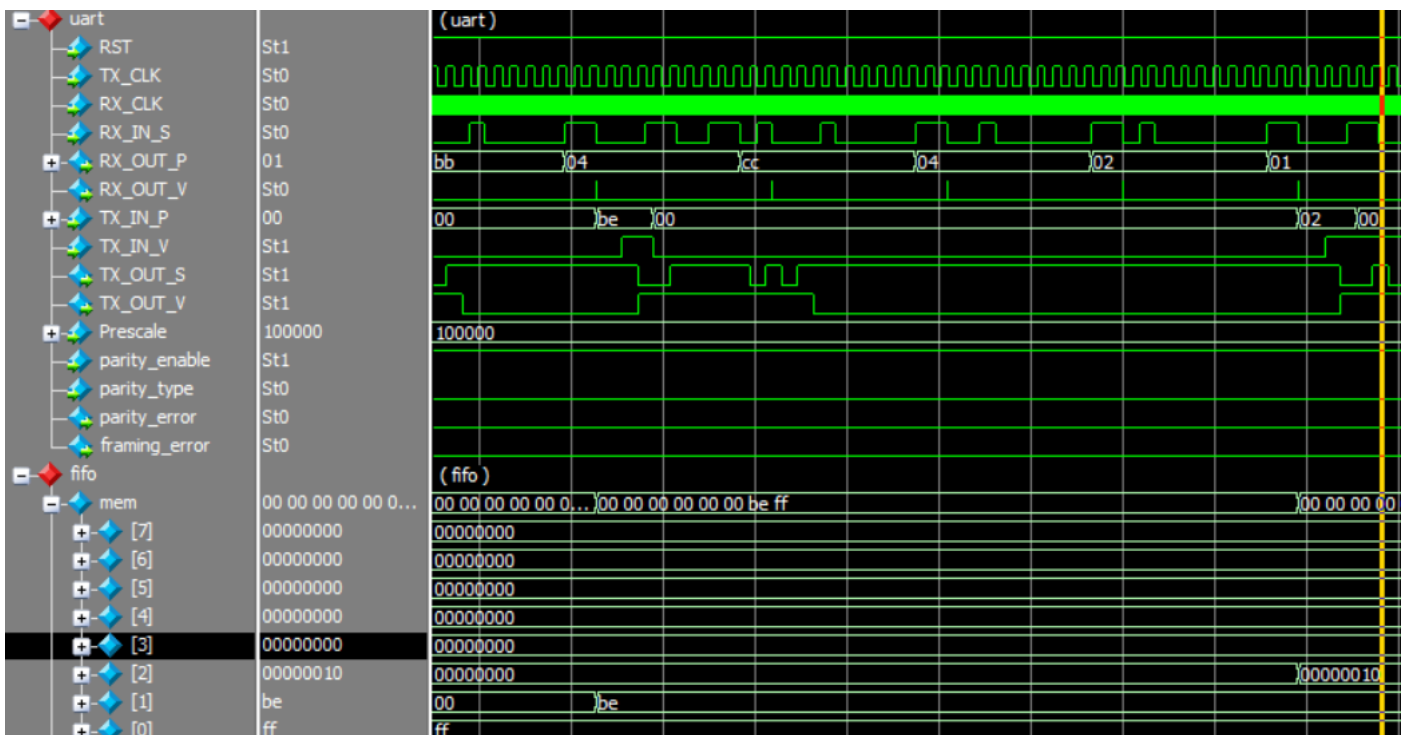
2. Register File Read command (2 frames)



Rx received frame 'bb' then the target address in the register file to read



Controller send it to the fifo and tx start sending it



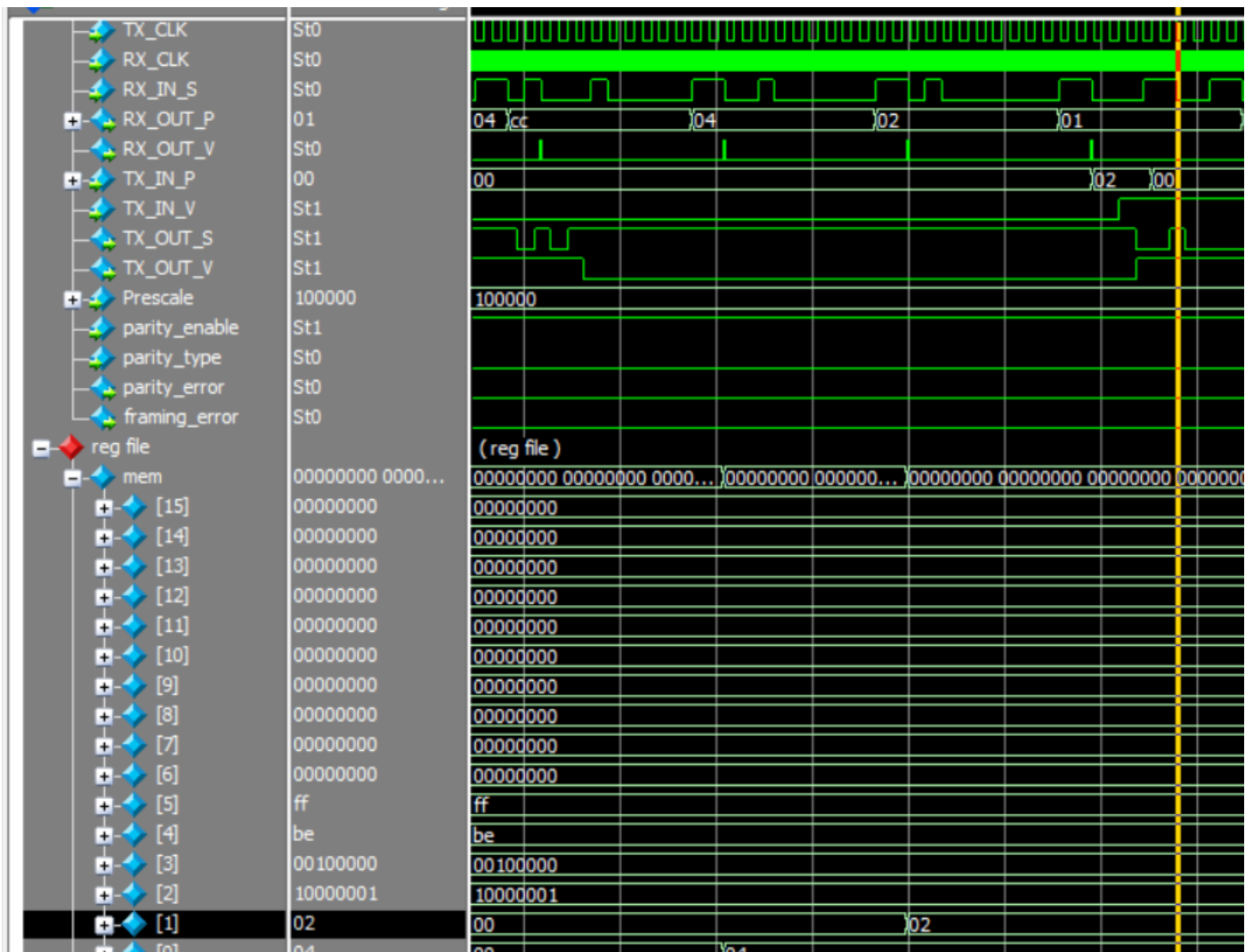
third case : A , B and function

3. ALU Operation command with operand (4 frames)



‘a’ and ‘b’ stored in register file address 0 , 1

With minus operation



Tx will send the output in two frames



fourth case : function only with old 'a' , 'b'

4. ALU Operation command with No operand (2 frames)



Rx received frame 'dd' then the operation which is division
Control unit take the result and send it in two frames as shown after the vertical yellow line

